Project Design Phase-II
Technology Stack (Architecture & Stack)

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2 **Architectural Diagram :-**

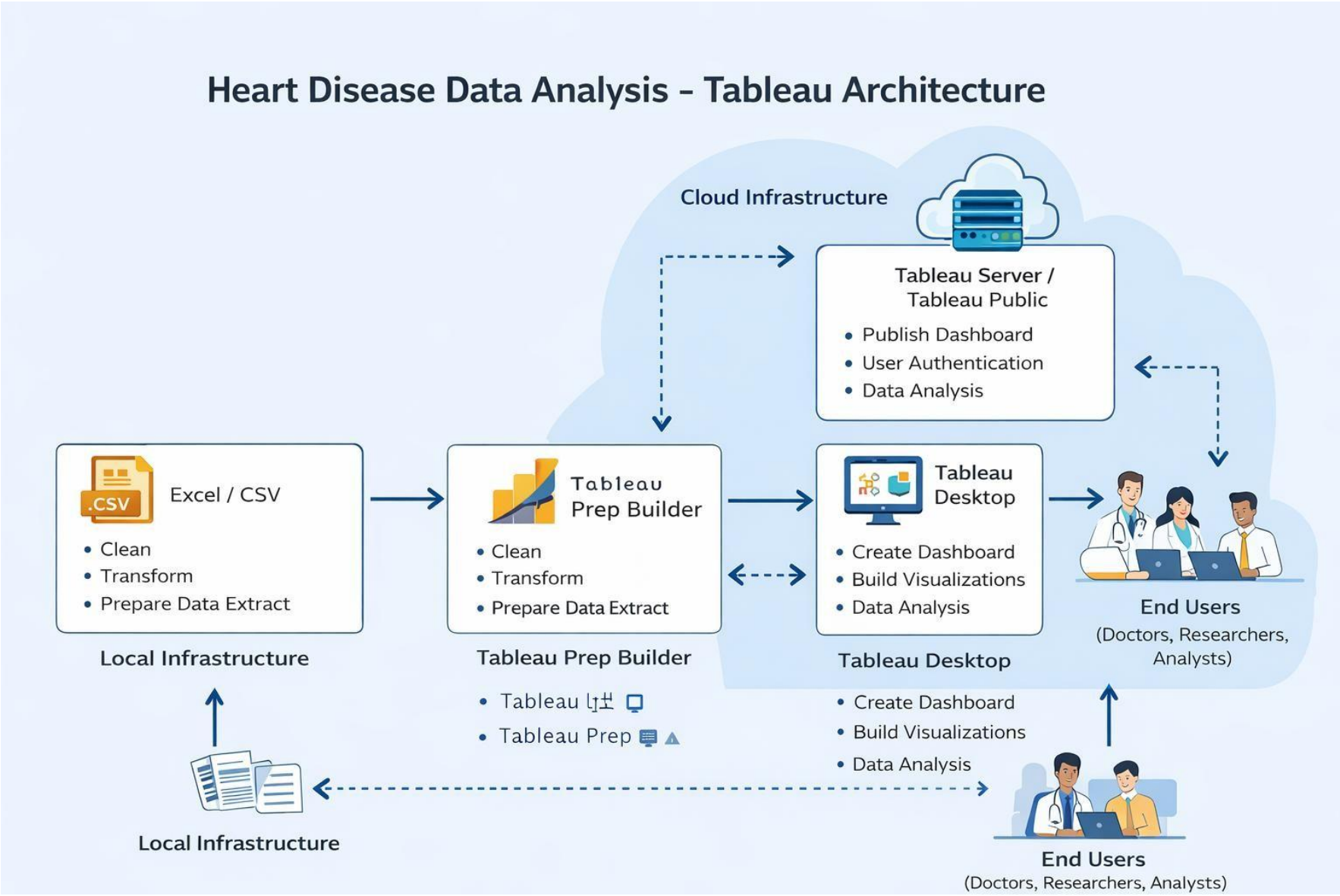

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interactive dashboards where users can explore heart disease data, filter patient attributes, and visualize risk factors through charts and graphs.	Tableau Dashboard (Tableau Desktop)
2	Application Logic-1	Data preparation process including cleaning, filtering, and structuring the heart disease dataset before visualization.	Tableau Prep Builder
3	Application Logic-2	Data transformation operations such as joins, unions, calculated fields, and aggregation of patient data.	Tableau Prep Builder
4	Application Logic-3	Data visualization and dashboard logic including filters, parameters, calculated fields, and interactive charts.	Tableau Desktop
5	Database	Source dataset containing patient health attributes like age, cholesterol, blood pressure, and heart disease indicators.	Excel / CSV Dataset
6	Cloud Database	Optional cloud storage used to host datasets for remote access and dashboard connectivity.	Tableau Server / Tableau Cloud
7	File Storage	Storage for datasets, Tableau workbook files (.twb / .twbx), and prepared data extracts.	Local File System
8	External API-1	Data connection interface used to import datasets from external files.	Tableau Data Connector
9	External API-2	Integration used to publish and share dashboards online.	Tableau Server / Tableau Public
10	Data Analysis Model	Analytical visualizations used to identify patterns, correlations, and risk factors in heart disease data.	Tableau Desktop

11	Infrastructure (Server / Cloud)	Platform used to publish and share dashboards with users.	Tableau Server / Tableau Public

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Tools used for visualization and data preparation in the project.	Tableau Desktop, Tableau Prep Builder
2	Security Implementations	Secure access to dashboards and datasets through authentication and role-based access control.	Tableau Server Security, User Authentication
3	Scalable Architecture	Dashboards and data extracts can handle large datasets and multiple user access.	Tableau Server Architecture
4	Availability	Dashboards can be accessed anytime through Tableau Server or Tableau Public.	Tableau Cloud / Tableau Public
5	Performance	Use of Tableau extracts and optimized dashboards ensures faster loading and interactive performance.	Tableau Data Extracts (.hyper), Dashboard Optimization